



ENGINEERS LEVEL · SEMESTER 3

Chatbot for Good

Student Workbook

Map inputs to helpful responses — and design AI that is honest, kind, and private.

Name: _____

THEORY

Chatbots, Prompts & Responsibility

A simple chatbot maps **intents** to **responses** using a **dictionary**. It reads **input**, finds a matching intent, and replies. **Prompt design** — clear, specific instructions — shapes what an AI produces. Responsible design: be honest that it's a bot, aim to genuinely help, and protect **privacy** (don't collect personal data).

Discuss: how should a helpful bot handle a question it cannot answer, and what personal data should it never ask for?

Trace it

Read the Python and give the response.

```
replies = { "hi": "Hello!", "help": "How can I help?" }
```

```
msg = "help"
```

```
print(replies[msg])
```

What is printed? _____

Code we use:

dict { }

input()

if / in

string methods

prompt design

function

Words we are learning

- **Dictionary** — { key: value } pairs, e.g. intent → response.
- **Input** — the user's message the bot reads.
- **Intent** — what the user is trying to do.
- **Prompt design** — writing clear instructions that shape AI output.
- **Ethics/privacy** — be honest it's a bot; don't collect personal data.

Name: _____ Date: _____

Dictionaries: Key → Value

A **dictionary** maps keys to values. Read it and add an entry.

```
replies = { "hi": "Hello!", "bye": "See you!" }
```

What is `replies["hi"]`? _____

Write a new key→value for a 'thanks' message:

```
"thanks": "_____"
```

Name: _____ Date: _____

Intents & Responses

Match each user message to its **intent** and a good response.

"what time is study group?"	intent: schedule → give the time
"I feel stressed"	intent: support → kind check-in + suggest a trusted adult
"tell me a joke"	intent: fun → a friendly joke

What should the bot do if it doesn't understand?

Name: _____ **Date:** _____

Prompt Design

Clear prompts get better results. Improve the weak prompt.

Weak: 'write something about study tips'

Rewrite it to be clear and specific (audience, length, focus):

Name: _____ **Date:** _____

Plan My Chatbot For Good

Plan a helpful bot and complete an ethics/privacy check.

Purpose (a real need it helps):

Three intents → responses:

Ethics/privacy check (honest it's a bot? any personal data? safe fallback?):

Name: _____ **Date:** _____

PROJECT

Chatbot for Good — Plan

Build a chatbot for a real need (study buddy, well-being check-in): map intents to responses with a dictionary, handle input, and pass an ethics/privacy check — honest, helpful, and no personal data collected.

1. The real need my bot helps with:

2. Intents dictionary:

```
replies = { _____ }
```

3. How it handles an unknown message:

4. Ethics/privacy safeguards:
